

Speech-Based Parkinson's Disease Analysis: From Acoustic Feature Engineering to Physics-Inspired Vision Transformers

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Abstract

Speech is a rich biological signal containing information about phonation, articulation, vocal-tract configuration, timing, and motor control. This paper combines a conventional speech-processing framework with a physics-inspired deep-learning architecture for Parkinson's disease (PD) analysis. The first part establishes the signal-processing foundation, including sampling, preprocessing, silence and noise reduction, voice activity detection, framing, pitch, jitter, shimmer, harmonics-to-noise ratio (HNR), formants, RMS energy, zero-crossing rate, spectral analysis, DFT/FFT, DCT, cepstral analysis, and MFCCs. It then considers classical machine learning, self-supervised learning, learned speech embeddings, LSTMs, and Transformers. The second part describes a time-frequency model in which a spectrogram is processed by a convolutional stem and Vision Transformer (ViT), with a CLS representation used for healthy-control (HC) versus PD classification. In parallel, a physics-inspired dynamical branch models a shared driving force $F(t)$ together with frequency-dependent damping $\gamma(f, t)$ and oscillatory parameters $\omega(f, t)$. Smoothness and gradient-difference regularization constrain the learned dynamics, while a Gradient Reversal Layer (GRL) encourages domain-invariant representations. The framework therefore connects interpretable acoustic descriptors, learned representations, temporal modeling, dynamical structure, and domain robustness.

Keywords: Parkinson's disease, speech processing, acoustic features, MFCC, self-supervised learning, spectrogram, Vision Transformer, physics-inspired learning, domain-adversarial learning, Gradient Reversal Layer.

1. Introduction

Speech is a highly informative biological signal. In addition to linguistic content, it contains information related to the physiological and neurological state of the speaker. Changes in vocal-fold control, respiratory coordination, articulatory movement, motor control, and speech timing can alter measurable acoustic properties. Parkinson's disease is associated with motor impairments, and speech may exhibit reduced loudness, monopitch, reduced intensity variation, imprecise articulation, altered timing, pauses, and vocal instability. These observations motivate the use of speech as a non-invasive signal for

computational PD analysis.

A conventional speech-learning pipeline can be summarized as

Speech $\rightarrow$ Preprocessing $\rightarrow$ Segmentation
 $\rightarrow$ Feature Extraction $\rightarrow$ ML/DL $\rightarrow$ Prediction.

The deep-learning perspective extends this to

Speech $\rightarrow$ Spectrogram $\rightarrow$ Encoder
 $\rightarrow$ Representation $\rightarrow$ Prediction.

while additionally introducing dynamical reconstruction and domain-adversarial branches. The two perspectives are complementary: hand-crafted features provide acoustic interpretability, whereas learned representations can capture complex patterns that are difficult to specify manually.

2. Speech Signal Representation

An analog speech signal may be represented by $x(t)$. After sampling,

$$x[n] = x(nT_s), \quad f_s = \frac{1}{T_s}.$$

For a recording of duration T , the approximate number of samples is

$$N = f_s T.$$

The Nyquist condition requires

$$f_{\max} < \frac{f_s}{2}.$$

Thus, a 16-kHz recording can represent frequencies up to approximately 8 kHz. Bit depth controls amplitude quantization; B bits provide 2^B possible quantization levels.

Raw recordings can contain silence, environmental noise, microphone noise, electrical interference, reverberation, speech, and non-speech events. A practical preprocessing sequence is

Raw recording $\rightarrow$ Silence removal $\rightarrow$ Noise reduction $\rightarrow$ Normalization

Normalization can be expressed as

$$x_{\text{norm}} = \frac{x - \mu}{\sigma}.$$

The normalization statistics should be computed using the appropriate training data to avoid information leakage.

3. Voice Activity Detection, Framing and Windowing

Voice Activity Detection (VAD) identifies speech-containing regions. An energy-based approach uses short-time RMS:

$$E_{\text{RMS}}[m] = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} x_m[n]^2}.$$

A frame may be classified as non-speech when

$$E_{\text{RMS}}[m] < \tau.$$

Zero-crossing rate is

$$\text{ZCR} = \frac{1}{N-1} \sum_{n=1}^{N-1} \mathbf{1}[x[n]x[n-1] < 0].$$

Energy and ZCR can be combined in practical VAD systems.

Speech is approximately stationary only over short intervals, so it is divided into overlapping frames:

$$x_m[n] = x[n + mH],$$

where H is the hop size. Windowing gives

$$x_w[n] = x_m[n]w[n].$$

For example, the Hamming window is

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right).$$

Windowing reduces discontinuities at frame boundaries and spectral leakage.

4. Traditional Acoustic Features

4.1 Fundamental Frequency and Pitch

The fundamental frequency F_0 describes the rate of vocal-fold vibration:

$$F_0 = \frac{1}{T_0}.$$

Pitch is the perceptual correlate of F_0 . Mean pitch is

$$\bar{F}_0 = \frac{1}{M} \sum_{m=1}^M F_0^{(m)},$$

and pitch variability is

$$\sigma_{F_0} = \sqrt{\frac{1}{M} \sum_{m=1}^M \left(F_0^{(m)} - \bar{F}_0\right)^2}.$$

Reduced pitch variability may be associated with mono-pitch in Parkinsonian speech.

4.2 Jitter and Shimmer

Jitter describes cycle-to-cycle variation in vocal period:

$$J_{\text{abs}} = \frac{1}{N-1} \sum_{i=1}^{N-1} |T_i - T_{i+1}|.$$

Shimmer measures amplitude variation:

$$S_{\text{abs}} = \frac{1}{N-1} \sum_{i=1}^{N-1} |A_i - A_{i+1}|.$$

Thus,

Jitter $\rightarrow$ frequency/period variation, Shimmer $\rightarrow$ amplitude variation

4.3 HNR, Formants and Energy

The harmonics-to-noise ratio is conceptually

$$\text{HNR}_{\text{dB}} = 10 \log_{10} \left(\frac{P_{\text{harmonic}}}{P_{\text{noise}}} \right).$$

Formants are vocal-tract resonances, commonly denoted $F_1, F_2, F_3, \dots$. They are affected by the configuration of the tongue, lips, jaw, and other vocal-tract structures.

RMS magnitude is

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{n=1}^N x[n]^2}.$$

Useful energy descriptors include mean RMS, RMS standard deviation, extrema, dynamic range, and temporal energy patterns.

5. Frequency-Domain Analysis and MFCCs

The Fourier transform expresses a signal in terms of frequency components:

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt.$$

For a discrete signal,

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}.$$

A direct DFT has approximately $O(N^2)$ complexity, whereas FFT algorithms reduce this to approximately $O(N \log N)$.

The DCT represents a sequence using cosine basis functions. It is particularly important for MFCCs because it approximately decorrelates filter-bank energies and concentrates information into a compact representation.

The MFCC pipeline is

Waveform $\rightarrow$ Framing $\rightarrow$ Windowing $\rightarrow$ FFT $\rightarrow$ Power Spectrum

A common mel-scale conversion is

$$m(f) = 2595 \log_{10} \left(1 + \frac{f}{700} \right).$$

For a filter $H_m[k]$ and power spectrum $P[k]$,

$$E_m = \sum_k P[k]H_m[k], \quad L_m = \log(E_m + \epsilon).$$

The DCT produces MFCC coefficients:

$$c_n = \sum_{m=1}^M L_m \cos \left[\frac{\pi n}{M} \left(m - \frac{1}{2} \right) \right].$$

A representative classical feature vector is

$$\left[F_0, \sigma_{F_0}, \text{Jitter}, \text{Shimmer}, \text{HNR}, \text{RMS}, \text{ZCR}, \right. \\ \left. F_1, F_2, F_3, \text{MFCC}_1, \dots, \text{MFCC}_K \right].$$

6. Classical and Learned Speech Representations

Classical machine-learning models operate on engineered acoustic features. Random Forest uses an ensemble of decision trees and majority voting:

$$\hat{y} = \text{mode}\{T_1(\mathbf{x}), \dots, T_B(\mathbf{x})\}.$$

An SVM seeks a decision boundary with a large margin. Feed-forward neural networks can also classify the acoustic feature vector.

Self-supervised learning (SSL) learns representations without requiring a manual label for every training example. A conceptual pipeline is

$$\begin{aligned} \text{Unlabeled speech} &\rightarrow \text{SSL pretraining} \\ &\rightarrow \text{Speech embedding} \\ &\rightarrow \text{PD classifier.} \end{aligned}$$

wav2vec-style systems can be summarized as

$$\begin{aligned} \text{Waveform} &\rightarrow \text{Feature encoder} \\ &\rightarrow \text{Context network} \\ &\rightarrow \text{Contextual representation.} \end{aligned}$$

A learned embedding such as $\mathbf{e} \in \mathbb{R}^{768}$ should not be interpreted as 768 independent hand-crafted acoustic features; its dimensions jointly form a learned representation.

Hand-crafted and learned features can also be combined:

$$\mathbf{h} = [\mathbf{e}; \mathbf{a}], \quad \hat{y} = f_\theta(\mathbf{h}).$$

7. Sequence Modeling

Speech is inherently sequential. LSTMs model temporal dependencies using recurrent gates, whereas Transformers use attention. Scaled dot-product attention is

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right) V.$$

Multi-head attention uses multiple projections:

$$\text{MHA}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O.$$

This permits relationships between distant speech frames to be modeled directly.

8. Physics-Inspired Spectrogram Representation

The deep-learning perspective begins with the spectrogram

$$X(f, t),$$

which describes acoustic energy over time and frequency. If there are 50 frequency bins, a time frame can be represented as

$$\mathbf{X}_t = [X(f_1, t), \dots, X(f_{50}, t)].$$

These 50 values are acoustic frequency bins, not 50 physical vocal-tract regions. Instead, articulator movement changes vocal-tract configuration, which changes acoustic resonances and therefore the distribution of energy over multiple frequency components:

$$\begin{aligned} \text{Articulator movement} &\rightarrow \text{Vocal-tract configuration} \\ &\rightarrow \text{Acoustic response} \\ &\rightarrow \text{Frequency distribution.} \end{aligned}$$

9. Physics-Inspired Dynamical Model

The acoustic dynamics are represented by the abstraction

$$\frac{d^2 X_f}{dt^2} + 2\gamma_f \frac{dX_f}{dt} + \omega_f^2 X_f = F.$$

Here, F represents a driving force or excitation, γ controls damping-related behavior, ω controls oscillatory or natural-frequency-related behavior, and X_f is the response associated with frequency component f .

The equation is a learned dynamical abstraction rather than a claim that every spectrogram bin is an independent physical oscillator.

The architectural assumption is

$$F(t) \rightarrow \text{shared excitation,}$$

while

$$\gamma(t) = [\gamma_1(t), \dots, \gamma_{50}(t)]$$

and

$$\omega(t) = [\omega_1(t), \dots, \omega_{50}(t)]$$

are frequency dependent. Thus, the 50 outputs correspond to acoustic frequency bins, not anatomical regions.

10. Dynamical Regularization

The learned force can become unnecessarily irregular. A smoothness penalty is

$$\mathcal{L}_{\text{Fsmooth}} = \sum_t |F_{t+1} - F_t|^2.$$

A gradient-difference penalty is

$$\mathcal{L}_{\text{grad}} = \sum_t [(F_{t+1} - F_t) - (F_t - F_{t-1})]^2$$

or equivalently

$$\mathcal{L}_{\text{grad}} = \sum_t (F_{t+1} - 2F_t + F_{t-1})^2.$$

The first controls first-order temporal variation, whereas the second controls changes in the temporal slope.

11. Convolutional Stem and Vision Transformer

The complete feature-extraction path is

Speech $\rightarrow$ Spectrogram $\rightarrow$ ConvStem
 $\rightarrow$ GELU $\rightarrow$ Stride Downsampling
 $\rightarrow$ Patch/Token Representation $\rightarrow$ ViT.

The convolutional stem extracts local time-frequency patterns and reduces resolution. Stride-based downsampling reduces the number of tokens and consequently the computational cost of self-attention. GELU is defined by

$$\text{GELU}(x) = x\Phi(x),$$

where $\Phi(x)$ is the standard Gaussian CDF.

A spectrogram is naturally two-dimensional, with time and frequency as its axes. It can therefore be divided into patches:

$$[P_1, P_2, \dots, P_N].$$

A CLS token is prepended:

$$[\text{CLS}, P_1, \dots, P_N].$$

After Transformer processing, h_{CLS} acts as a global representation.

For binary classification,

$$h_{\text{CLS}} \rightarrow \text{Linear Layer} \rightarrow [P(\text{HC}), P(\text{PD})].$$

The HC/PD prediction is produced by this main classification head.

12. Domain-Adversarial Learning

A domain can represent systematic acquisition differences such as datasets, microphones, recording environments, or sessions. The desired representation should rely on speech characteristics relevant to the task rather than nuisance domain characteristics.

The representation $\mathbf{z}$ is sent to

$$\mathbf{z} \rightarrow \text{HC/PD classifier}, \quad \mathbf{z} \rightarrow \text{GRL} \rightarrow \text{Domain classifier}.$$

During the forward pass,

$$\text{GRL}(\mathbf{z}) = \mathbf{z}.$$

During backpropagation,

$$\frac{\partial \text{GRL}}{\partial \mathbf{z}} = -\lambda \mathbf{I}.$$

Thus, if the domain classifier receives gradient $\mathbf{g}$, the encoder receives $-\lambda \mathbf{g}$.

The domain classifier attempts to identify the domain, while the feature extractor is encouraged to make domain identification difficult. A conceptual objective is

$$\min_{\theta_f, \theta_g} \max_{\theta_d} [\mathcal{L}_{\text{class}} - \lambda \mathcal{L}_{\text{domain}}].$$

Importantly, GRL does not classify HC versus PD and does not assign relevance probabilities to individual features. The final HC/PD prediction comes from the main classification head.

13. Stochastic Depth and Learning-Rate Scheduling

Stochastic depth randomly skips complete residual blocks during training. Unlike dropout, which masks individual activations, stochastic depth removes an entire residual path for a training iteration.

The cosine learning-rate schedule is

$$LR_t = LR_{\min} + \frac{1}{2}(LR_{\max} - LR_{\min}) \left[1 + \cos \left(\frac{\pi t}{T} \right) \right].$$

It allows larger updates early in training and smaller refinements later. The scheduler controls the learning rate rather than directly changing the definition of the loss.

14. Reconstruction, Prediction and Causality

Reconstruction can be used as a representation-learning objective:

$$\text{Speech} \rightarrow \text{Encoder} \rightarrow \text{Latent representation} \rightarrow \text{Reconstruction}.$$

The learned representation can subsequently support supervised classification. This is particularly useful when labeled data are limited.

Causality must be distinguished from positional encoding. A causal model uses only permitted past information, whereas a non-causal model may use future context:

$$x_t \leftarrow x_{t-k}, \dots, x_t, \dots, x_{t+k}.$$

Causality is imposed by the attention pattern, typically using a mask that prevents attention to future positions. Positional encoding itself is not causal.

A standard sinusoidal positional encoding is

$$PE(pos, 2i) = \sin \left(\frac{pos}{10000^{2i/d_{\text{model}}}} \right),$$

$$PE(pos, 2i+1) = \cos \left(\frac{pos}{10000^{2i/d_{\text{model}}}} \right).$$

15. Multi-Objective Training

A conceptual total objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}} + \lambda_{\text{grad}} \mathcal{L}_{\text{grad}} + \lambda_{\text{adv}} \mathcal{L}_{\text{domain}}.$$

The domain component is implemented adversarially through GRL rather than simply minimized by the feature extractor. The exact coefficients depend on the final implementation.

For ordinary binary classification, binary cross-entropy may be written as

$$\mathcal{L}_{\text{BCE}} = -[y \log p + (1-y) \log(1-p)].$$

Table 1: Example experimental comparison.

Exp.	Representation	Classifier
1	MFCC only	SVM
2	Classical acoustic features	Random Forest
3	Classical acoustic features	MLP
4	Learned speech embedding	MLP
5	Learned speech embedding	SVM
6	Classical + learned	MLP
7	Frame sequence	LSTM
8	Frame sequence	Transformer

16. Evaluation and Experimental Design

Evaluation should not rely on accuracy alone. Relevant confusion-matrix metrics include precision, recall/sensitivity, specificity, F1 score, ROC curves, and AUC.

The experimental design should also control subject-level leakage. In PD analysis, subject-independent splitting is essential because recordings from the same speaker should not unintentionally appear in both training and test sets.

This comparison addresses whether learned representations provide information beyond carefully designed acoustic features.

17. Integrated End-to-End Framework

The combined research framework can be summarized as

Speech waveform $\rightarrow$ Preprocessing + VAD
 $\rightarrow$ Spectrogram
 $\rightarrow F$ -head, γ -head, ω -head
 $\rightarrow$ ConvStem + GELU + Downsampling
 $\rightarrow$ Patch embeddings
 $\rightarrow [\text{CLS}, P_1, \dots, P_N]$
 $\rightarrow$ Vision Transformer
 $\rightarrow h_{\text{CLS}}$.

The learned representation can feed multiple branches:

$$h_{\text{CLS}} \rightarrow \begin{cases} \text{HC/PD classification head,} \\ \text{GRL} \rightarrow \text{domain classifier,} \end{cases}$$

This architecture combines four complementary forms of information. Conventional acoustic descriptors provide interpretable measurements of phonation and articulation. Learned embeddings provide high-dimensional representations capable of capturing complex patterns. The ViT models long-range time-frequency relationships. The physics-inspired and adversarial branches impose structural biases: the former models temporal dynamics, while the latter suppresses domain-specific nuisance information.

18. Discussion

The traditional and deep-learning perspectives should not be viewed as competing descriptions of speech.

MFCCs, pitch, jitter, shimmer, HNR, formants, RMS, and ZCR explicitly encode recognizable acoustic properties. Learned embeddings, by contrast, can encode combinations of information distributed across many dimensions. A hybrid system can therefore preserve interpretability while exploiting learned representations.

The physics-inspired branch adds a different inductive bias. It does not claim that each frequency bin corresponds to an anatomical oscillator. Rather, it uses the damped-system equation as a structured abstraction of how acoustic components evolve over time. The shared $F(t)$ represents common excitation, while $\gamma(f, t)$ and $\omega(f, t)$ allow frequency-specific temporal behavior.

The GRL branch has a separate purpose. It is not a relevance estimator and does not produce the final disease prediction. Its purpose is to alter the training gradient so that the encoder retains task-relevant information while reducing information that makes recording domains easy to identify.

19. Limitations and Practical Considerations

Several sources of error should be considered. Recording noise, microphone differences, room acoustics, speaker variability, class imbalance, and data leakage can all produce misleading results. Aggressive denoising may also remove acoustic characteristics that are useful for diagnosis. Similarly, a complex model does not automatically provide a stronger experiment than a carefully controlled classical baseline.

The distinction between offline analysis and true real-time prediction is also important. A non-causal Transformer can exploit both past and future context, whereas a strict real-time system cannot use future frames beyond its allowed latency. Therefore, claims of causality or real-time operation must be tied to an explicit operational definition.

20. Conclusion

This paper combines a comprehensive speech-processing foundation with a physics-inspired Transformer architecture for Parkinson’s disease analysis. The signal-processing framework establishes how speech is sampled, cleaned, segmented, transformed into frequency-domain representations, and converted into interpretable acoustic features such as pitch, jitter, shimmer, HNR, formants, RMS, ZCR, and MFCCs. Classical machine learning, self-supervised learning, learned embeddings, LSTMs, and Transformers provide progressively more flexible ways of modeling these representations.

The deep-learning framework represents speech as a spectrogram, uses a convolutional stem to learn local acoustic patterns, and applies a Vision Transformer to capture global relationships between time-frequency patches. The CLS representation supports the final HC/PD prediction. A physics-inspired dynamical branch introduces a shared excitation $F(t)$ and frequency-dependent $\gamma(f, t)$ and $\omega(f, t)$, while smoothness losses regularize the learned trajectory. A GRL-

based domain-adversarial branch encourages robustness to systematic recording-domain differences. Stochastic depth and cosine learning-rate scheduling provide additional optimization and regularization strategies.

The resulting progression is

Speech $\rightarrow$ Acoustic/Spectral Representation $\rightarrow$ Learned Representation $\rightarrow$ Structured Dynamics $\rightarrow$ Robust Classification.

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